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Md Muhtasim Munif Fahim
University of Rajshahi
Md. Rezaul Karim
mrkarim@ru.ac.bd

University of Rajshahi

Article

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Cost-Aware Evaluation of Time-Series Foundation Models for Urban Air-Quality and Temperature Forecasting

Md Muhtasim Munif Fahim[1] **and Md. Rezaul Karim**[1,*]

[1]Data Science Research Lab, Department of Statistics, University of Rajshahi, Rajshahi-6205, Bangladesh
[*]Corresponding author: Md. Rezaul Karim (mrkarim@ru.ac.bd)

ABSTRACT

Cities with the greatest need for hyper-local air-quality and weather forecasts often have the shortest data records and the tightest compute budgets. The usual responses, searching for a tiny architecture or transferring from data-rich cities, assume foundation models are too heavy for such deployments, an assumption adopted far more often than tested. Across 29 cities and two domains (hourly PM2.5 and 2 m temperature), under pre-specified protocols with measured energy and a cost-adjusted decision rule, we compare all three. On PM2.5 the zero-shot model and the tuned specialist are statistically indistinguishable (MASE 0.662 vs 0.692 under deployable causal covariates; formally equivalent at a 0.05-MASE margin only when the specialist is granted perfect-foresight covariates). The specialist's apparent temperature edge (0.533 vs 0.792) shrinks to a small, non-significant difference (0.745) once its covariates are restricted to values knowable at forecast time: a perfect-foresight artifact that can produce spurious cross-domain conclusions in any benchmark using future covariates. Transfer learning never significantly beats zero-shot at any budget. Under frequent retraining the untrained model uses roughly ten times less measured energy, though a once-trained specialist eventually amortizes. Where models are refreshed often, downloading one now matches training one.

Keywords: time-series foundation models, air-quality forecasting, temperature forecasting, neural architecture search, transfer learning, edge computing, energy-aware machine learning, deployment cost

Introduction

Ambient fine particulate matter is among the largest environmental risk factors for human health, responsible for millions of premature deaths annually and concentrated disproportionately in cities of low- and middle-income countries[1,2]. Managing that burden (issuing health advisories, timing traffic or industrial interventions, planning sensor maintenance) requires hyper-local, short-horizon forecasts at the level of an individual monitor. The cities that need such forecasts most are also those where monitoring is newest and sparsest. Low-cost sensor networks and open platforms such as OpenAQ have widened coverage[3,4], but the records they produce are short, gappy, and heterogeneous, and the institutions operating them rarely command cloud budgets or ML engineering teams[5]. The same constraint binds for local weather variables such as near-surface temperature, whose station-level forecasts feed heat-warning and energy systems[6]: global deep-learning weather models now rival numerical weather prediction (NWP)[7,8], but their outputs and compute remain out of reach of a single-station operator on an edge budget.

A resource-constrained operator today faces three deployment strategies. The first is to train an efficient specialist locally. Gradient-boosted trees remain a strong widely deployable baseline for tabularized forecasting[9–11], deep specialists from recurrent probabilistic models to graph and transformer architectures are widely represented in the air-quality and general forecasting literatures[12–19], and neural architecture search (NAS) refines the recipe by looking for an architecture that meets an explicit efficiency budget[20–22]. The second is transfer learning: pretrain where data are plentiful, then fine-tune on the short local record[23]. In our prior conference study we combined the two routes: we discovered a 153k-parameter recurrent forecaster by multi-objective search and showed that transfer from data-rich cities recovers most of the accuracy lost to short local histories[24]. The present work inherits that architecture and its edge/energy framing, and adds the foundation-model tier, the cost-adjusted decision rule, the air-quality domain, and the transfer-versus-zero-shot crux experiment. The third strategy, available only since 2023, is to train nothing at all: pretrained time-series foundation models (TSFMs) such as Chronos, TimesFM, Moirai, Lag-Llama and TimeGPT forecast zero-shot from the local context window alone[25–31].

Which strategy should such an operator choose? The benchmark literature does not settle the question. Foundation-model evaluations report aggregate leaderboards across heterogeneous datasets[32], and recent critiques question how robust the

zero-shot advantage is[33]; the doubt echoes much older evidence that complex methods do not automatically beat simple ones[34]. Deployment cost is largely absent from these comparisons, although the green-AI literature has long argued that energy belongs in the objective[35–39] and the edge-ML community treats the joule budget as the binding constraint[40,41]. Underlying much of that work, including our conference study above[24], is the assumption that foundation models are too heavy for the edge and that efficiency must therefore come from search and transfer. The assumption is reasonable, and it has been adopted far more often than it has been tested. Compact foundation models make the test feasible: Chronos-Bolt-small runs on a consumer GPU[42], so what has served as a design premise becomes a question the data can answer.

A methodological problem compounds the strategy question. Forecast evaluation has well-documented pitfalls[43,44], and we examine one that matters specifically for comparisons across domains. Standard evaluation harnesses hand covariate-using models their exogenous inputs at the forecast target time (the `future_covariates` convention), which quietly assumes a perfect weather forecast over the horizon. The assumption is harmless where covariates carry little information about the target and decisive where they are near-deterministic drivers of it, and we show below that it can, by itself, produce a spurious domain-level conclusion about which strategy wins.

Here we test the three strategies directly. We compare five tiers on a 29-city panel (14 data-rich, 15 data-scarce) in two domains, urban PM2.5 and location-level temperature: a seasonal-naïve floor, a LightGBM specialist, the NAS-discovered Green-NAS-A model, zero-shot Chronos-Bolt, and a covariate-residual Chronos variant. Evaluation uses pre-specified quality gates, six-fold rolling-origin backtesting, and Diebold–Mariano, sign, and Wilcoxon tests under false-discovery-rate (FDR) control, with paired equivalence tests wherever we claim a tie; uncertainty is quantified with split-conformal intervals, and energy is measured directly on edge/CPU-class hardware (a consumer GPU and a CPU-only configuration). A pre-specified crux experiment (E4) pits the transfer-learning recipe against the zero-shot model across fine-tune budgets on the scarce cities. A cost-adjusted decision rule, minimizing the mean absolute scaled error (MASE) + $\lambda \cdot$ USD per 1,000 forecasts with λ a cost-penalty coefficient, then converts the results into deployment winner maps.

The paper makes three contributions.

The first is the test that the premise had not received. To our knowledge this is the first head-to-head comparison of search, transfer, and zero-shot deployment for hyper-local urban forecasting in which accuracy, measured energy, and cost enter a single decision rule, on a panel large enough to separate the data-rich case from the data-scarce one. The premise does not survive it. On PM2.5, zero-shot Chronos-Bolt and the tuned specialist are statistically indistinguishable across the panel: neither records an FDR-significant win over the other, a paired two-one-sided equivalence test (TOST) declares the two equivalent at a 0.05-MASE margin when the specialist is granted perfect-foresight covariates, and under deployable causal covariates the foundation model is, if anything, ahead. Its advantage shows no detectable association with local data volume. In the data-scarce regime, transfer learning never significantly beats the zero-shot model at any fine-tune budget, and the zero-shot model leads at every one.

The second is methodological and outlives this dataset: covariate timing alone can manufacture a domain-level conclusion. On temperature, the specialist's large apparent superiority is an artifact of the standard future-covariates convention. Once covariates are restricted to values knowable at the forecast origin, its lead shrinks to a small, non-significant margin, and the corrected cross-domain picture is graded rather than domain-flipped. Because the distortion scales with how informative the covariates are about the target, it is nearly invisible within one domain and surfaces only in comparisons across domains, which is exactly where it does the most damage. Any benchmark that consumes future covariates and compares across domains needs a causal-covariate ablation to rule it out.

The third is the deployment consequence, which is what an operator can act on. Under a frequent-retraining protocol the zero-shot model uses roughly an order of magnitude less measured energy than the tiers that train locally. Separating training from inference shows that gap is training-driven rather than intrinsic: a once-trained specialist eventually amortizes and becomes the more energy-efficient option, so the defensible recommendation is conditional on refresh cadence rather than universal. For a city that retrains often, lacks a trustworthy forecast of its own covariates, or has no ML engineering capacity to spare, the practical default has moved from training a small model to downloading one. We release the decision rule so that an operator can re-derive that answer at their own electricity price, hardware, and retraining frequency.

Results

Study design

Five model tiers in two domains, urban PM2.5 (primary) and 2 m air temperature (secondary), on one 29-city panel under one pre-specified harness (Fig. 1; Methods). The tiers span the three options open to a resource-constrained operator: *search* (NAS-GRU, the architecture carried over from our earlier conference study[24]), *specialize* (LightGBM with weather and calendar covariates[9]), and *train nothing* (Chronos-Bolt-small run zero-shot, with and without a covariate-residual variant[25,42]), plus a seasonal-naïve floor. Accuracy is MASE on a fixed per-series scale; energy is measured end to end[45]. Results below are at the

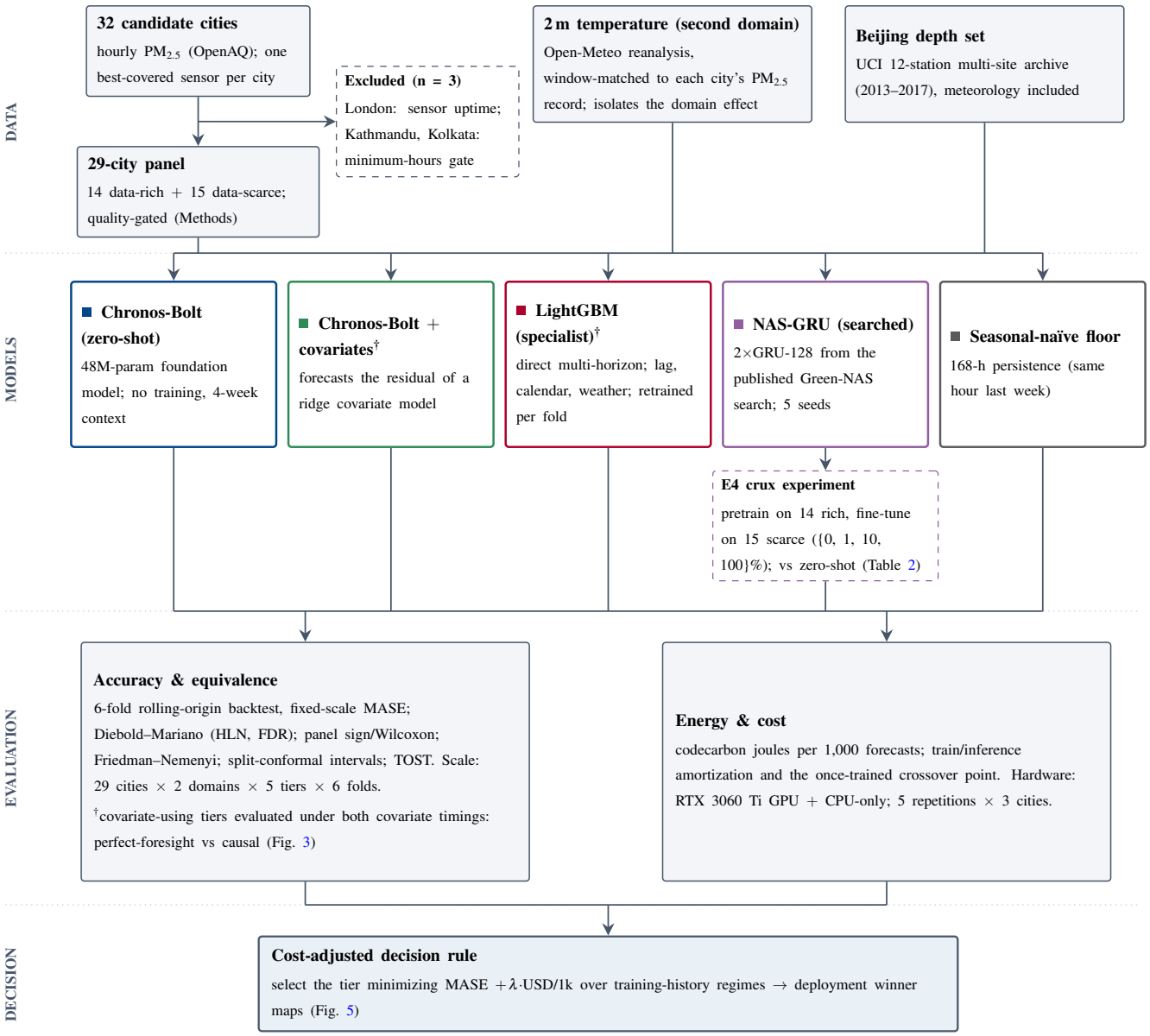


Figure 1. Study design: five forecasting strategies, two domains, 29 cities. Of 32 candidate cities, 3 were excluded under pre-specified quality criteria, leaving a 29-city panel (14 data-rich, 15 data-scarce). Hourly PM2.5 (OpenAQ, one reference sensor per city) and 2 m temperature (Open-Meteo, window-matched to each city's PM2.5 record) are forecast 24 h ahead by five tiers spanning search (NAS-GRU, with the E4 transfer experiment), specialization (LightGBM direct multi-horizon), zero-shot foundation modelling (Chronos-Bolt-small, with and without a covariate-residual variant), and a seasonal-naïve floor; daggers mark the covariate-consuming tiers evaluated under both covariate timings. Every tier is scored on identical six-fold rolling-origin backtests with fixed-scale MASE, Diebold–Mariano/sign/Wilcoxon inference under FDR control, split-conformal intervals, and TOST equivalence tests, and separately on codecarbon-measured energy; the cost-adjusted decision rule then combines accuracy and cost into deployment winner maps.

24-hour horizon; the 48-hour horizon behaves the same way (Supplementary Table S9). Abbreviations used throughout are collected in Supplementary Table S1.

A zero-shot foundation model matches the tuned specialist on PM2.5

On the 29-city PM2.5 panel, zero-shot Chronos-Bolt and the tuned LightGBM specialist were statistically indistinguishable. In the deployable causal-covariate configuration reported in Table 1, panel-mean MASE is 0.662 ± 0.368 (Chronos-Bolt) versus 0.692 ± 0.374 (specialist) (Fig. 2a); with perfect-foresight covariates the specialist matches Chronos-Bolt exactly

Table 1. Panel accuracy by domain, covariate configuration, and tier. MASE, MAE, and RMSE (mean ± sd across 29 cities; NAS-GRU collapsed to per-city seed-means first; MAE/RMSE in target units, μg m^{-3} for PM2.5 and °C for temperature), Friedman–Nemenyi average rank across the five tiers (lower is better; critical difference 1.13 at $\alpha = 0.05$, $k = 5$, $n = 29$), and FDR-corrected Diebold–Mariano significant-win counts against zero-shot Chronos-Bolt (tier wins / foundation-model wins). Each domain carries two blocks: the deployable *causal* configuration (weather covariates at their last value known at the forecast origin; the primary result, as in Figs. 2a and 3) and the *perfect-foresight* upper bound (covariates at the forecast target time; per-city detail in Supplementary Table S13). The univariate zero-shot and naïve tiers use no weather covariates, so their accuracy is identical in both blocks; their rank still shifts because competitors move. Under causal covariates no tier records an FDR-significant win over the zero-shot model in either domain, and the specialist–foundation-model rank gap is inside the critical difference in both domains (PM2.5: 2.41 vs 1.72; temperature: 1.72 vs 1.93); the perfect-foresight temperature block shows the 6-win specialist upper bound that the causal ablation removes.

Domain	Covariates	Tier	MASE (mean ± sd)	MAE (mean ± sd)	RMSE (mean ± sd)	Avg. rank	FDR wins (tier / FM)
$PM_{2.5}$	Causal (deployable)	Chronos-Bolt (zero-shot)	0.662 ± 0.368	10.553 ± 13.524	15.093 ± 19.273	1.72	—
		Chronos-Bolt + covariates	0.797 ± 0.481	13.500 ± 19.643	18.425 ± 26.624	3.34	0 / 5
		LightGBM (specialist)	0.692 ± 0.374	10.902 ± 13.617	15.245 ± 19.246	2.41	0 / 0
		NAS-GRU	0.734 ± 0.352	11.426 ± 14.931	15.592 ± 20.791	2.90	0 / 2
		Seasonal-naïve	1.026 ± 0.530	16.505 ± 21.235	22.274 ± 28.983	4.62	0 / 13
	Perfect foresight	Chronos-Bolt (zero-shot)	0.662 ± 0.368	10.553 ± 13.524	15.093 ± 19.273	1.83	—
		Chronos-Bolt + covariates	0.730 ± 0.376	11.745 ± 15.266	15.959 ± 20.478	3.14	1 / 2
		LightGBM (specialist)	0.662 ± 0.371	10.845 ± 14.060	14.810 ± 18.895	2.17	0 / 0
		NAS-GRU	0.734 ± 0.352	11.426 ± 14.931	15.592 ± 20.791	3.07	0 / 2
		Seasonal-naïve	1.026 ± 0.530	16.505 ± 21.235	22.274 ± 28.983	4.79	0 / 13
Temperature	Causal (deployable)	Chronos-Bolt (zero-shot)	0.792 ± 0.282	1.303 ± 0.717	1.819 ± 1.040	1.93	—
		Chronos-Bolt + covariates	2.614 ± 1.044	3.923 ± 1.821	4.919 ± 2.217	4.72	0 / 27
		LightGBM (specialist)	0.745 ± 0.223	1.242 ± 0.701	1.629 ± 0.903	1.72	0 / 1
		NAS-GRU	0.999 ± 0.481	1.626 ± 1.093	2.070 ± 1.261	2.59	0 / 6
		Seasonal-naïve	1.686 ± 1.050	2.716 ± 1.780	3.300 ± 2.085	4.03	0 / 16
	Perfect foresight	Chronos-Bolt (zero-shot)	0.792 ± 0.282	1.303 ± 0.717	1.819 ± 1.040	2.90	—
		Chronos-Bolt + covariates	0.718 ± 0.278	1.228 ± 0.795	1.559 ± 1.012	2.41	1 / 2
		LightGBM (specialist)	0.533 ± 0.208	0.888 ± 0.550	1.153 ± 0.730	1.28	6 / 0
		NAS-GRU	0.999 ± 0.481	1.626 ± 1.093	2.070 ± 1.261	3.59	0 / 6
		Seasonal-naïve	1.686 ± 1.050	2.716 ± 1.780	3.300 ± 2.085	4.83	0 / 16

(0.662, Supplementary Table S13). Neither model recorded a single FDR-corrected DM-significant win over the other across all 29 cities, in either the causal panel (0/0) or the perfect-foresight panel (0/0) (full perfect-foresight pairwise matrices in Supplementary Table S5). In the perfect-foresight configuration the panel-level tests were non-significant (sign test P = 0.136, Wilcoxon signed-rank P = 0.325). Under causal covariates the per-city balance in fact tilts toward the foundation model, which is better in 21 of 29 cities (sign test P = 0.024), but the magnitude-weighted test does not reach significance (Wilcoxon P = 0.084). We therefore describe the deployable comparison conservatively, as no significant difference rather than a foundation-model win.

A non-significant test is not evidence of equivalence, so we tested equivalence directly. Under perfect foresight the agreement clears that stricter bar: a paired two-one-sided test against a margin of $\delta = 0.05$ MASE declares the perfect-foresight specialist *equivalent* to Chronos-Bolt (TOST P = 0.026; the 90% interval that TOST requires at $\alpha = 0.05$, [−0.042, 0.042], lies inside the margin). Under causal covariates the point difference is +0.030 MASE and non-significant, but the interval is a little too wide to certify equivalence at the same margin (upper bound 0.071), so we report that comparison as "no significant difference" only. All seven equivalence tests appear in Supplementary Fig. S2 and their sensitivity to δ in Supplementary Table S15. Both models cleared the seasonal-naïve floor (naïve MASE 1.026 ± 0.530; 13 FDR-significant Chronos-Bolt wins over naïve). The NAS-searched model and the covariate-residual Chronos variant trailed (0.734 ± 0.352 and 0.797 ± 0.481), and a Friedman test across all five tiers rejected tier equivalence overall (P < 0.001 in both configurations). The tie is therefore specific to the specialist-versus-foundation-model comparison; the leaderboard as a whole is not flat.

The foundation model's advantage showed no detectable correlation with how much local history a city had: in the deployable causal configuration, the correlation between usable hours and the specialist-minus-FM MASE gap was r = −0.031 (P = 0.873, bootstrap 95% CI [−0.287, 0.251]), and the perfect-foresight panel is likewise flat (r = 0.075, P = 0.698). Zero-shot competitiveness is therefore not a scarce-data phenomenon that longer records would erase. This is a cross-sectional check across heterogeneous cities and is read descriptively; usable hours is a coarse proxy, since the zero-shot model reads a fixed

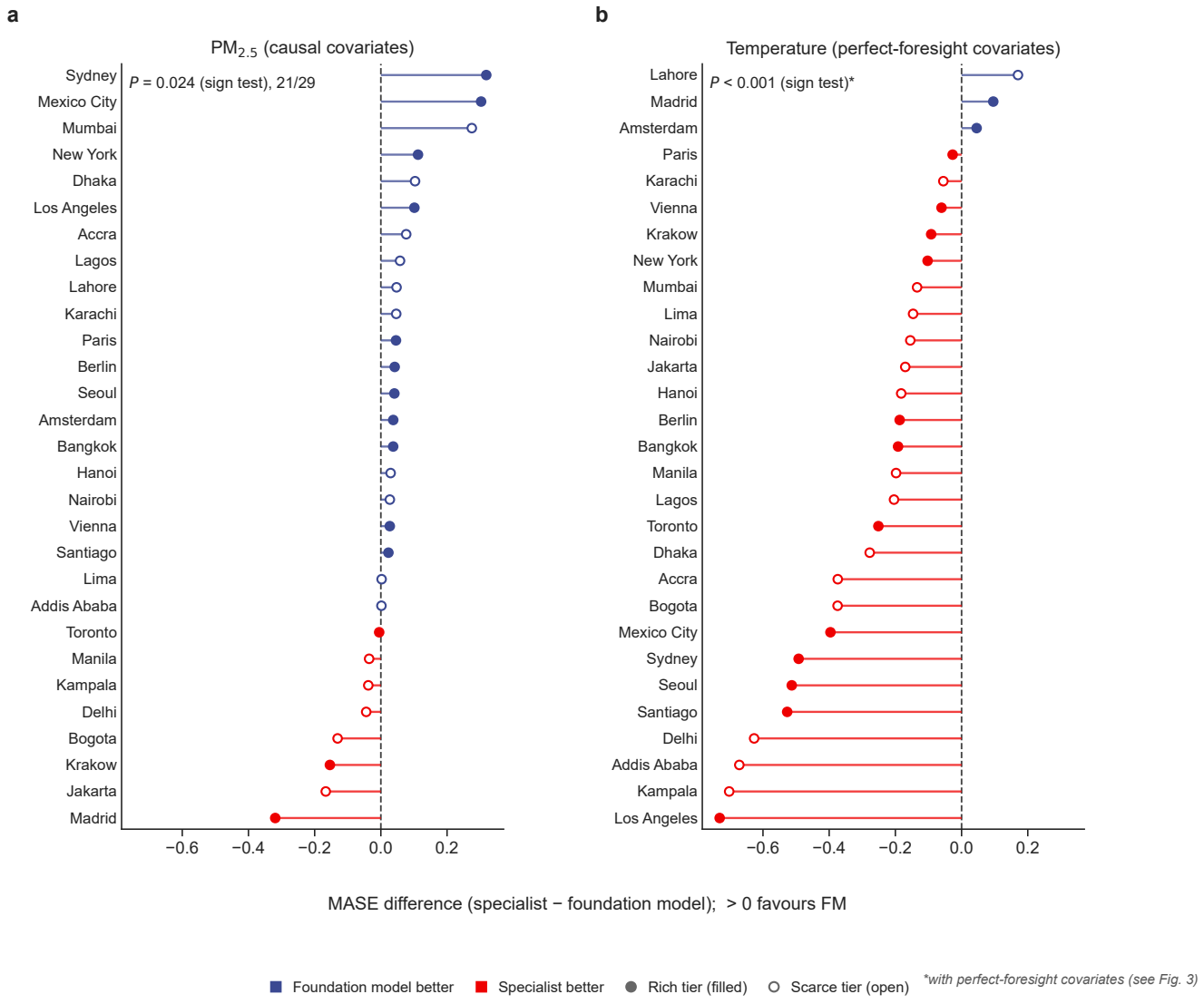


Figure 2. Zero-shot foundation model versus tuned specialist across 29 cities. Per-city MASE difference (specialist − foundation model; positive favours the foundation model), sorted within panel; marker fill distinguishes data-rich from data-scarce cities ($n = 29$ cities per panel). (**a**) PM2.5 under the deployable causal covariates of Table 1: the foundation model is better in 21 of 29 cities (sign test $P = 0.024$), but no per-city difference is FDR-significant and the Wilcoxon test does not reach significance ($P = 0.084$), so there is no significant difference overall. (**b**) 2 m temperature at face value favours the specialist; this panel uses perfect-foresight covariates (the `future_covariates` convention; sign test $P < 0.001$), and Fig. 3 shows the lead disappears under causal covariates.

four-week context while trained tiers can exploit the full history.

The specialist's temperature edge requires a perfect weather forecast

At face value the temperature domain looked like a specialist win. With perfect-foresight covariates (reported as an upper bound in Supplementary Table S13), LightGBM reached panel-mean MASE 0.533 ± 0.208 versus 0.792 ± 0.282 for zero-shot Chronos-Bolt, with 6 FDR-significant specialist wins and none in the other direction (sign and Wilcoxon P < 0.001) (Fig. 2b). That comparison, however, is a perfect-foresight one: it hands every covariate-using tier the weather at the forecast *target* time (Methods). For PM2.5 the assumption adds little, because weather is a weak 24-h predictor of pollution. For temperature the covariates (humidity, pressure, radiation, dew point) are near-deterministic physical drivers of the target, so the assumption does a great deal of work.

We therefore re-ran the specialist, the strongest covariate user and the tier carrying the domain claim, under causal covariates.

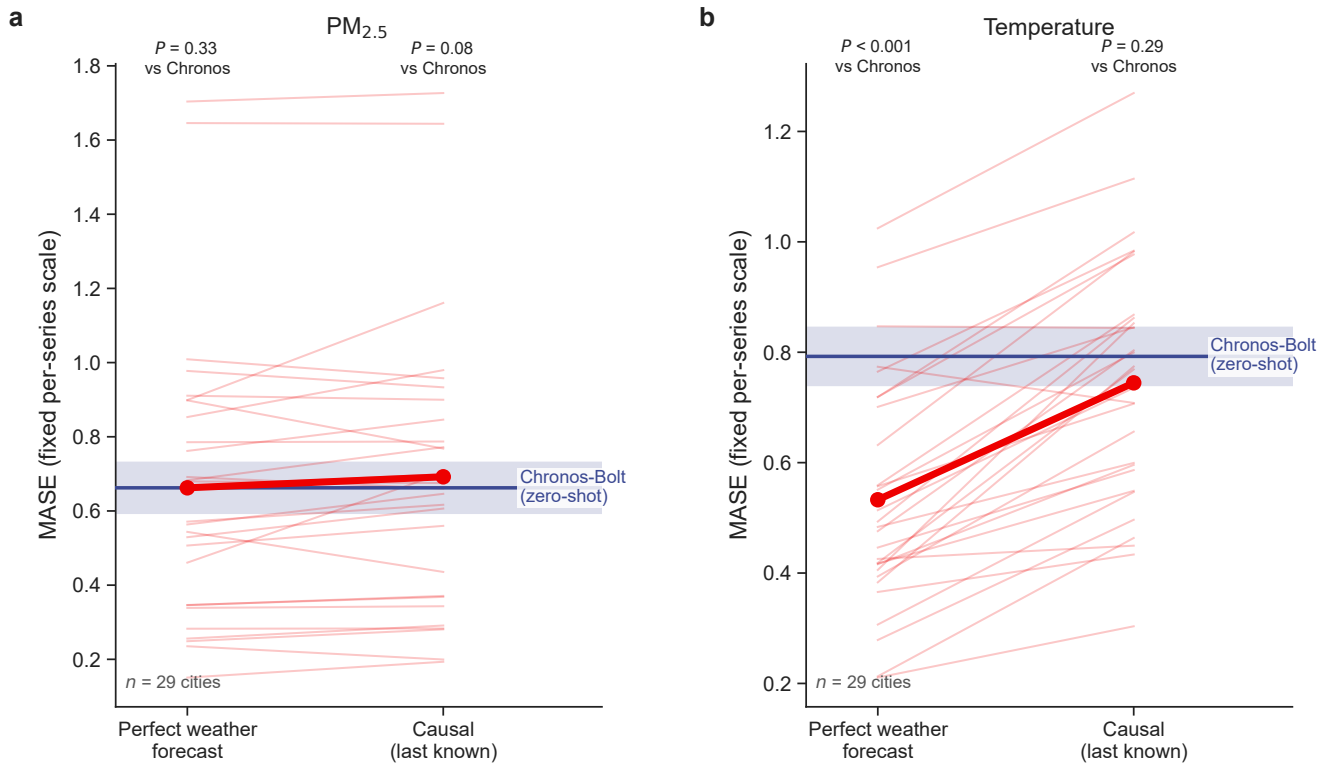


Figure 3. The specialist's temperature edge requires a perfect weather forecast. Per-city LightGBM MASE with perfect-foresight covariates (left ticks) versus causal, last-known covariates (right ticks), $n = 29$ cities per panel; bold line = panel mean; horizontal band = zero-shot Chronos-Bolt panel mean ± s.e.m. across cities. On-panel P-values are paired Wilcoxon signed-rank tests of LightGBM against Chronos-Bolt. (**a**) PM2.5: foresight is worth ∼0 (0.662 vs 0.692; both statistically indistinguishable from the foundation model). (**b**) Temperature: removing foresight moves the specialist from 0.533 (26/29 city wins, $P = 1.4 \times 10^{-6}$) to 0.745 (16/29, $P = 0.29$, not significant) against the foundation model at 0.792.

The restriction removed the temperature lead (Fig. 3). Panel-mean MASE rose from 0.533 to 0.745, the per-city win rate against Chronos-Bolt fell from 26/29 (Wilcoxon P = 1.4×10^{-6}) to 16/29 (P = 0.29, no longer significant), and the foresight assumption alone was worth +0.212 MASE (P = 2.6×10^{-8}). In the causal-covariate panel that Table 1 now reports as the main result, the specialist records zero FDR-significant DM wins over the foundation model, and the foundation model records one (0/1), a reversal of the 6/0 count seen under perfect foresight. The same restriction hits the covariate-residual Chronos variant far harder in this domain, because its ridge component extrapolates stale covariates across the horizon. Its panel-mean MASE degrades to 2.614 ± 1.044, worse than the seasonal-naïve floor (Table 1). The same ablation left PM2.5 unmoved (0.662 perfect-foresight vs 0.692 causal, both statistically indistinguishable from Chronos-Bolt at 0.662; P = 0.33 and P = 0.08). Held to the stricter equivalence standard, the two domains differ only in which direction the residual uncertainty leans. On PM2.5, formal equivalence at the 0.05-MASE margin is established in the perfect-foresight configuration (above); under causal covariates it is not, and the interval leaves room only on the specialist's unfavourable side (90% CI [−0.012, 0.071]). On temperature under causal covariates equivalence is likewise not established, but the interval leans the specialist's way: the paired difference favours it by a small, non-significant margin (0.745 vs 0.792; 90% CI [−0.114, 0.018]). The corrected cross-domain finding is therefore not a domain flip but a graded one. Under realistic covariate availability, the small zero-shot foundation model shows no statistically significant accuracy deficit to the tuned specialist in either domain (formal equivalence is established on PM2.5 when the specialist is granted perfect-foresight covariates), and the specialist's large apparent weather edge is an artifact of assuming a perfect forecast of its own inputs. Perfect-foresight numbers are retained in all tables as a clearly-labelled upper bound on specialist performance. As in PM2.5, the temperature-domain FM advantage showed no detectable linear association with data volume in the causal configuration (Pearson r = 0.138, P = 0.477, 95% CI [−0.220, 0.452]). With only 29 cities this is an absence of detectable association, not evidence of true independence.

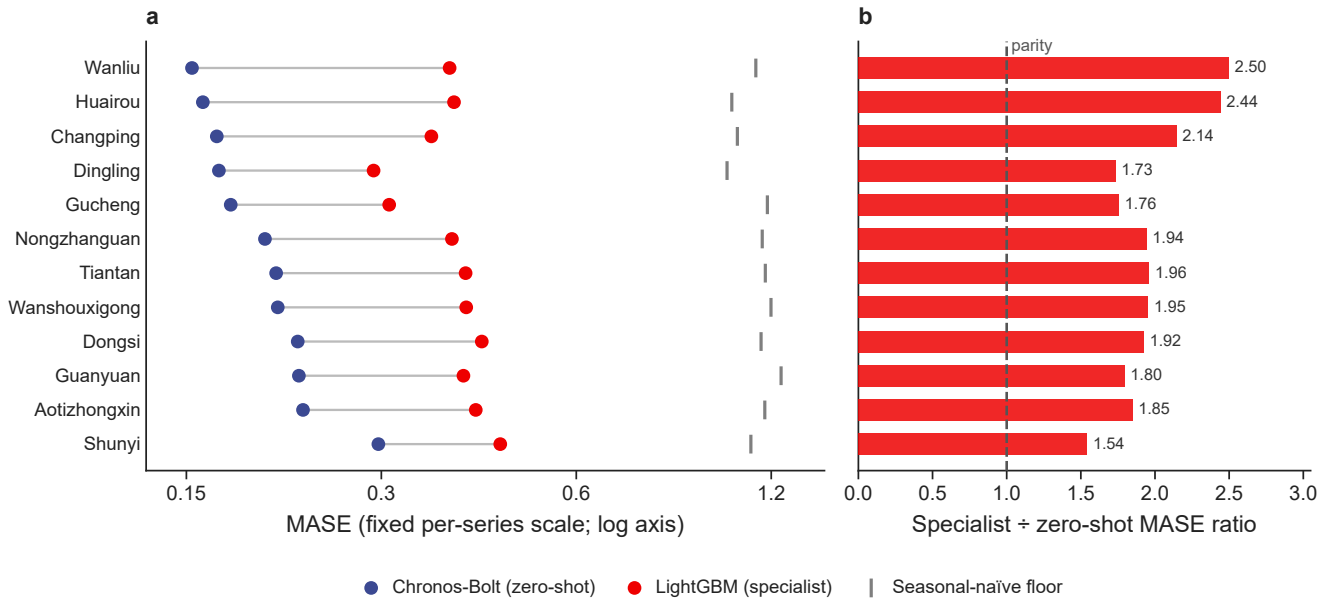


Figure 4. Beijing multi-station depth check. (**a**) Per-station MASE at the 12 UCI Beijing multi-site stations (six-fold backtest, after the pre-specified value-sanitization step; Methods). Zero-shot Chronos-Bolt (0.153–0.297) beats the LightGBM specialist (0.292–0.458) at 12/12 stations; the seasonal-naïve floor spans 1.03–1.24. (**b**) Specialist-to-foundation-model MASE ratio per station (same station order); the dashed line marks parity. The advantage is uniform across the metropolitan network rather than driven by a subset of stations, so the panel-level tie is not an averaging artifact.

Beijing multi-station depth check

One sensor per city trades depth for breadth, so we added a 12-station within-city check[46]. Zero-shot Chronos-Bolt beat the LightGBM specialist at 12 of 12 stations, and beat both the specialist and the seasonal-naïve floor at all 12 (Chronos-Bolt MASE range 0.153–0.297; LightGBM 0.292–0.458; naïve 1.03–1.24) (Fig. 4). The panel-level tie is therefore not an averaging artifact over heterogeneous stations.

The evaluated transfer recipe does not outperform zero-shot in the data-scarce case

The crux question inherited from our earlier conference study[24] is whether a NAS-discovered model *pretrained on data-rich cities and fine-tuned on a scarce city* still beats a foundation model that has never seen the target city. It does not. Across the 15 scarce-tier cities at fine-tune budgets of {0, 1, 10, 100}% of local history (Methods), zero-shot Chronos-Bolt held the best mean MASE at every budget: 0.843 versus 0.899/0.915/0.888/0.876 for transferred NAS-GRU and 0.941/0.944/0.858 for a LightGBM refit on the same budget (Table 2). The transfer recipe never overtakes zero-shot: after Holm correction across fractions, the city-level paired Wilcoxon tests find no fraction at which transfer significantly beats zero-shot. The paired difference in fact favours the zero-shot model at every budget, nominally significantly at the 0% and 1% budgets (bootstrap 95% CI on the difference excludes zero) and non-significantly at 10% and 100%. Equivalence is not established at any budget (all TOST 90% CIs exceed the 0.05-MASE margin; Supplementary Fig. S2). We are therefore not claiming that the two are interchangeable, only that the previously published transfer recipe does not recover a data-scarce advantage over simply running the foundation model. Strategy choice in the scarce regime therefore turns on cost and operational simplicity, and there the zero-shot model wins (next subsection).

Energy, cost, and the deployment decision

Under our backtesting-and-retraining protocol, the measured end-to-end energy reverses the expectation that the foundation model is the expensive option (Table 3). Because the specialist retrains one model per lead time per fold while the zero-shot model only runs inference, LightGBM consumed 9.5–15.1 kJ per 1,000 forecasts across the three replication cities, against 1.0–1.2 kJ for zero-shot Chronos-Bolt on a consumer GPU (one repeatability-flagged Nairobi cell ranged up to 5.4 kJ). That is roughly an order of magnitude in energy, and about 8× in cost (6.9×10^{-4} versus 8.2×10^{-5} USD per 1,000 forecasts at 0.15 USD/kWh and a power-usage-effectiveness (PUE) factor of 1.4). NAS-GRU, which must be trained locally, sits with the specialist (7.3–16.8 kJ/1k). Cells that failed the repeatability gate are shown as ranges rather than point estimates throughout (4 of 15; Methods).

Table 2. Transfer learning versus zero-shot across fine-tune data budgets (E4). Panel results over the 15 data-scarce cities: NAS-GRU pretrained on the 14 data-rich cities and fine-tuned at nominal budgets of {0, 1, 10, 100}% of local history (5 seeds, per-city seed-means first), and a LightGBM specialist refit on the same nominal budget (its window floors at 202 h, so no 0% row exists; the asymmetric data floors at the 1% budget are described in Methods). Columns: mean actual training hours behind each nominal budget, panel MASE (mean ± sd across cities), paired panel-mean difference versus zero-shot Chronos-Bolt (positive means zero-shot is better), per-city win counts (strategy / foundation model), Holm-corrected paired Wilcoxon P across budgets, and the TOST equivalence interval on the paired difference for the pre-specified transfer comparisons (computed for the confirmatory NAS-GRU family). Two quantities in this table share the value 0.05 but are unrelated: the equivalence margin $\delta = 0.05$ is measured in MASE units, whereas the significance level is $\alpha = 0.05$. The reported interval is a 90% one because TOST tests against the $1 - 2\alpha$ interval, not the 95% interval; equivalence at $\delta = 0.05$ is not established at any budget. Zero-shot Chronos-Bolt holds the best panel mean at every budget, and no transfer budget significantly beats it (all Holm $P \geq 0.19$). The LightGBM refits are significantly worse at the two smallest budgets (Holm P = 0.009 at the 1% budget and 0.006 at the 10% budget), but not at the 100% budget.

Strategy	Budget (%)	Hours (mean)	MASE (mean ± sd)	Δ vs. zero-shot (panel mean)	Wins (strat. / FM)	Holm P (Wilcoxon)	TOST 90% CI ($\delta = 0.05$)
Chronos-Bolt (zero-shot)	—	—	0.843 ± 0.354	—	—	—	—
NAS-GRU (transfer + fine-tune)	0	0	0.899 ± 0.340	+0.056	4 / 11	0.192	[+0.005, +0.107]
	1	66	0.915 ± 0.341	+0.072	5 / 10	0.250	[+0.015, +0.129]
	10	599	0.888 ± 0.350	+0.045	7 / 8	0.421	[-0.012, +0.102]
	100	5,995	0.876 ± 0.383	+0.033	4 / 11	0.338	[-0.009, +0.075]
LightGBM (refit on budget)	1	202	0.941 ± 0.311	+0.098	1 / 14	0.009	—
	10	599	0.944 ± 0.383	+0.100	1 / 14	0.006	—
	100	5,995	0.858 ± 0.382	+0.015	6 / 9	0.421	—

That comparison reflects a frequent-retraining deployment, not a train-once one, and the distinction matters. Separating one-time training energy from per-forecast inference energy shows the measured gap is almost entirely a training gap. At inference alone, the specialist's 24 small per-horizon tree models are cheaper per forecast than a single Chronos-Bolt forward pass, in all three replication cities on CPU and in the two cities with a valid GPU estimate (the Beijing GPU inference measurement is a sub-kilojoule artifact; Supplementary Table S12). A specialist that is trained once and then reused therefore crosses over to being the more energy-efficient option after roughly 2,800–4,400 forecasts on CPU (≈ 120–180 days of hourly 24-h-ahead forecasting for one series) and 250–2,100 on GPU. The order-of-magnitude energy advantage of the zero-shot model is thus specific to deployments that retrain frequently (more often than every few months); it is not a universal property. The electricity cost of either tier is below 10^{-3} USD per 1,000 forecasts, so in absolute monetary terms the difference is small next to hardware, memory, and engineering costs. The zero-shot model's durable advantages are avoiding the training step altogether and its operational simplicity, not the electricity bill (Discussion).

Folding cost into the objective produces deployment winner maps across training-history regimes for one depth city (Beijing) plus one rich (Seoul) and one scarce (Nairobi) panel city in both domains (Fig. 5). Each cell selects the tier minimizing MASE + $\lambda \cdot$ USD/1k over the sweep of the cost-penalty coefficient λ (Methods): $\lambda = 0$ selects on accuracy alone, and larger λ penalizes cost more heavily. The main maps use the deployable causal covariates; a perfect-foresight version is a labelled upper bound (Supplementary Fig. S1). In PM2.5 the two configurations are nearly identical (weather is barely informative there) and the foundation-model family wins most of the causal map: 21/25 cells in Beijing, 14/25 in Seoul, and 13/20 in Nairobi to the Chronos family (8 zero-shot + 5 covariate variant). In temperature the configurations diverge sharply. Under perfect foresight the specialist takes many cells (11/25 Beijing, 12/25 Seoul, 18/20 Nairobi), but under causal covariates its share collapses: the Chronos family and NAS-GRU take most cells (Beijing 15/25 Chronos vs 6/25 specialist; Seoul 25/25 NAS-GRU; Nairobi 11/20 Chronos plus 8/20 NAS-GRU), consistent with the ablation. The maps are robust to the cost assumptions at the central setting and degrade gracefully away from it: across the full electricity-price × PUE sweep, no cells flip at the central assumption and at most 40% flip at the most extreme corners (Supplementary Table S11).

Uncertainty calibration

Split-conformal 95% intervals were near nominal for all tiers in both domains (Supplementary Table S4; per-city coverage and width in Supplementary Tables S6–S7). In the causal-covariate panel, pooled empirical coverage ranged 0.910–0.970 on PM2.5 and 0.900–0.968 on temperature; the low end is the weather rich-tier zero-shot pool at 0.90 against the 0.95 nominal. Under these realistic covariates the specialist's temperature intervals (rich-tier mean width 9.6 °C-scale units) were only marginally narrower than zero-shot Chronos-Bolt's (10.7), and most of the sharpness the perfect-foresight configuration gave it (7.7,

Table 3. Measured energy, cost, and model size per tier and replication city. codecarbon-measured joules per 1,000 forecasts for every tier in each of the three replication cities (mean ± sd over five repetitions; min–max range shown where sd/mean > 0.20), per-city cost at 0.15 USD/kWh with PUE 1.4 (the decision rule's central assumptions), the per-city CPU train/inference decomposition (one-time training energy per fit, per-forecast inference energy, and the crossover, meaning the forecasts a once-trained specialist serves before its amortized energy per forecast falls below the foundation model's; measured only for the two tiers in the amortization protocol, GPU rows in Supplementary Table S12), and parameter count as a hardware-independent size proxy (LightGBM's tree ensemble has no directly comparable unit and is not reported). Measured J/1k figures include per-fold training for trained tiers; the zero-shot tiers run inference only. The covariate-residual variant measures below plain zero-shot because tiers run sequentially within each measured workload and the zero-shot tier, which runs first, absorbs the one-time GPU initialization and model-load cost. Per-forecast latency and peak memory were not logged for the canonical runs.

Tier	City	Measured (J/1k)	Cost (USD/1k)	Train (J/fit, CPU)	Inference (J/forecast, CPU)	Crossover (forecasts, CPU)	Params
Chronos-Bolt (zero-shot)	Beijing	1,229 ± 241	7.2×10^{-5}	—	0.112	—	47.72M
	Seoul	1,025 ± 141	6.0×10^{-5}	—	0.110	—	
	Nairobi	1,056–5,417	1.1×10^{-4}	—	0.111	—	
Chronos-Bolt + covariates	Beijing	763 ± 102	4.5×10^{-5}	—	—	—	47.72M
	Seoul	781 ± 94	4.6×10^{-5}	—	—	—	
	Nairobi	842 ± 109	4.9×10^{-5}	—	—	—	
LightGBM (specialist)	Beijing	15,061 ± 2,374	8.8×10^{-4}	317	0.033	4,041	—
	Seoul	10,832 ± 406	6.3×10^{-4}	246	0.022	2,778	
	Nairobi	9,461 ± 836	5.5×10^{-4}	203	0.065	4,416	
NAS-GRU	Beijing	9,367–16,788	7.7×10^{-4}	—	—	—	157,080
	Seoul	7,496 ± 440	4.4×10^{-4}	—	—	—	
	Nairobi	7,298 ± 105	4.3×10^{-4}	—	—	—	
Seasonal-naïve	Beijing	2 ± 0	1.2×10^{-7}	—	—	—	—
	Seoul	1–36	5.3×10^{-7}	—	—	—	
	Nairobi	1–40	9.3×10^{-7}	—	—	—	

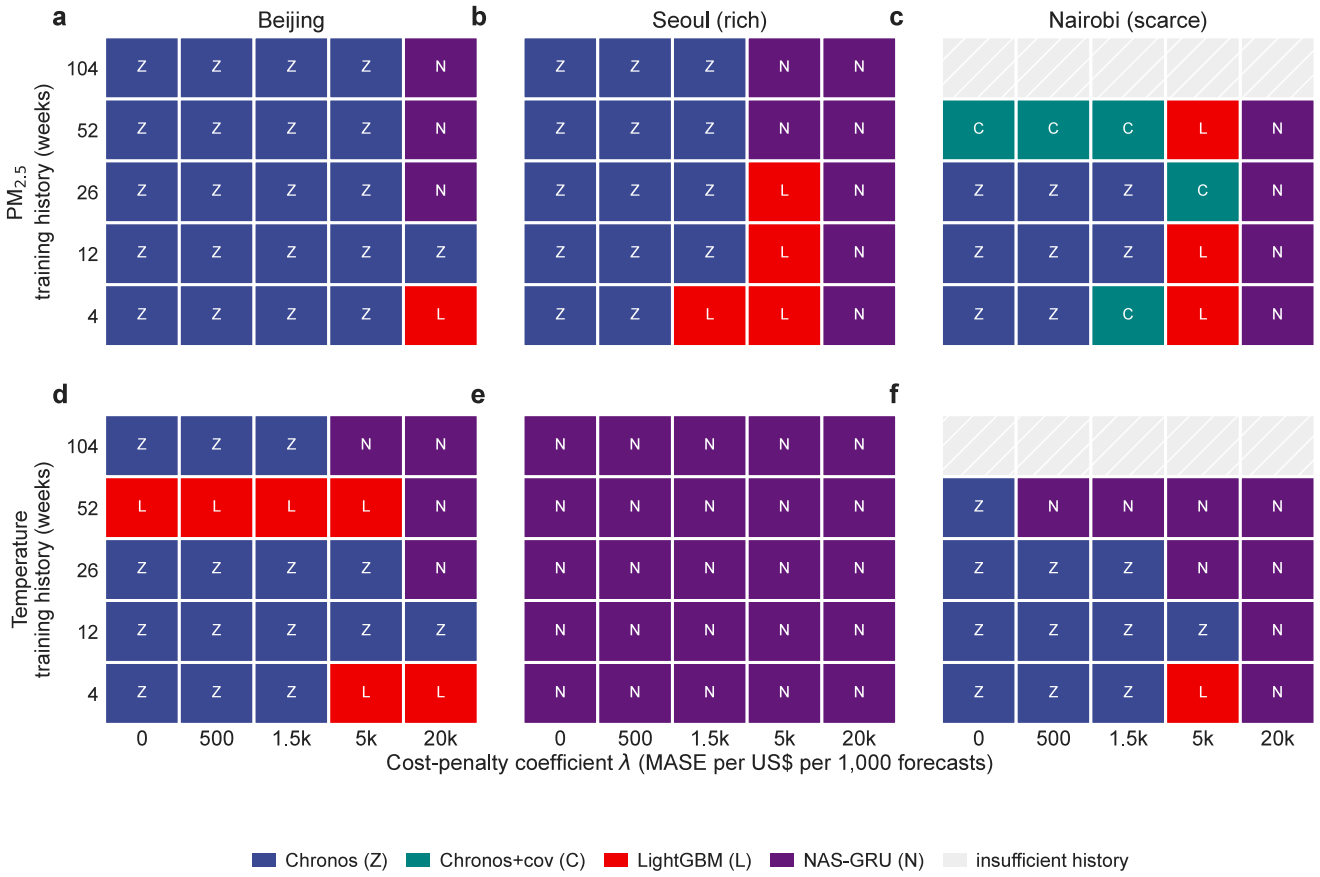


Figure 5. Cost-adjusted deployment winner maps (causal covariates). Winning tier under the objective MASE $+\lambda\cdot$ USD/1,000 forecasts, per training-history regime (rows: weeks of history) and cost-penalty coefficient λ (columns), for Beijing, Seoul, and Nairobi in both domains (**a**–**f**), using the deployable causal (last-known) covariates. PM2.5 maps are dominated by the Chronos family (21/25, 14/25, and 13/20 cells). Under causal covariates the temperature maps no longer favour the specialist: the Chronos family and NAS-GRU take most cells (e.g. Beijing 15/25 Chronos vs 6/25 specialist; Seoul 25/25 NAS-GRU), a marked shift from the perfect-foresight version (Supplementary Fig. S1), consistent with the ablation (Fig. 3). Sweeping electricity price over 0.05–0.30 USD/kWh and PUE over 1.0–2.0 flips no cells at the central assumption and at most 40% at the most extreme corners (Supplementary Table S11). Cell letters duplicate the colour coding for grayscale readers.

Supplementary Table S4) is lost. On PM2.5 the foundation model's intervals were in fact tighter than the specialist's (rich-tier 18.5 vs 21.0). The naïve floor achieved coverage only through much wider intervals (35.1 vs ∼18 on rich-tier PM2.5).

Discussion

The efficiency calculus has changed

The premise of our earlier conference study was that foundation models are too computationally heavy for edge deployment, so data-scarce cities need searched architectures and transfer learning[24]; the assumption was widely shared at the time and, there as elsewhere, was carried as a design constraint rather than measured. This study keeps the same evaluation philosophy and supplies the missing comparison. With a compact foundation model in the panel, the constraint no longer binds. Under realistic covariate availability, zero-shot Chronos-Bolt is statistically indistinguishable from the tuned specialist on PM2.5 (and equivalent by a formal TOST when the specialist is given perfect covariates), no longer significantly worse on temperature, and never significantly beaten by the transfer recipe at any fine-tune budget, where it leads at every one. Under a frequent-retraining protocol it also uses roughly an order of magnitude less energy, because it never trains.

Why should a model that has never seen a city forecast it about as well as a model trained there? A plausible reading is that the quantities a local specialist must estimate from a short, gappy record (the shape of the diurnal cycle, weekly structure, the persistence of pollution episodes) are the same generic regularities a foundation model has already absorbed from large,

heterogeneous pretraining corpora[25], so a four-week context window suffices to localize them. This reading is consistent with our observation that the zero-shot advantage shows no detectable association with local record length. It also sharpens, rather than contradicts, recent skepticism about foundation-model leaderboards[32,33]. In our panel the zero-shot model does not beat a well-tuned gradient-boosting specialist, which extends the long record of simple methods matching complex ones[34]; but matching the specialist *without training* is precisely what changes the deployment decision. The earlier work's comparisons remain internally sound, and search remains the right tool when an application needs a bespoke model; what changed is that small TSFMs[25,42] opened a second route to the same destination and moved the accuracy–cost frontier with it. "Make it tiny by search" and "make it cheap by not training" now land in the same accuracy band, and the second is operationally simpler. For an operator in a data-scarce city, the practical default shifts from training a small model to downloading one.

A covariate-timing pitfall that can produce spurious domain conclusions

A cross-domain claim can be an artifact of covariate timing alone. Had we used the standard `future_covariates` convention uncritically, this paper would have reported a clean domain flip (foundation model wins PM2.5, specialist wins temperature), with 26/29 significant city wins backing the second half. The causal ablation shows that result depends entirely on granting the specialist a perfect forecast of its own near-deterministic drivers: with last-known covariates the specialist's lead is no longer statistically significant (16/29 cities, P = 0.29), shrinking from +0.259 to +0.047 MASE, though a paired equivalence test does not certify the two as equivalent in this domain. The asymmetry is domain-dependent (worth +0.212 MASE in temperature and $\sim$0 in PM2.5) because covariate informativeness about the target differs by domain, and that is exactly what makes the artifact hard to detect in cross-domain comparisons: the same convention, applied uniformly, distorts one domain and not the other. Forecast-evaluation guides catalogue leakage and backtest pitfalls[43,44]; we add covariate foresight to that list and recommend that benchmarks using future covariates report a causal-covariate ablation alongside, or at minimum bracket reality between the two configurations. An operator with access to NWP-grade covariate forecasts sits between our causal floor and perfect-foresight ceiling[6]; our data show how wide that bracket can be.

Cost-aware deployment guidance

The winner maps turn the statistics into a decision, and for an operator that decision reduces to three questions our measurements can answer.

How much local history is there? Less than one might expect is enough to matter. At the shortest regime we map, four weeks of training history, the zero-shot model already takes most cells, and the size of its advantage shows no detectable association with record length across the panel ($r = -0.031$, $P = 0.873$ on PM2.5 under causal covariates). Over the full causal maps the Chronos family takes 21 of 25 cells in Beijing, 14 of 25 in Seoul and 13 of 20 in Nairobi. An operator choosing on accuracy alone, at realistic covariate availability, has little reason to train.

Is there a trustworthy forecast of the covariates? This is the condition under which a specialist clearly earns its place, and it is the one an operator is most likely to get wrong. Granted perfect-foresight weather, the specialist takes many temperature cells (11/25 Beijing, 12/25 Seoul, 18/20 Nairobi); restricted to what is knowable at the forecast origin, that share collapses. Most operators sit between those two cases, and the width of the bracket, worth +0.212 MASE in temperature and about zero in PM2.5, measures directly how much an NWP-grade covariate feed[6] is worth before it is worth training a specialist to consume it.

How often will the model be refreshed? This, rather than the electricity price, decides which option is cheaper. Under frequent retraining the zero-shot model uses roughly an order of magnitude less energy and about eight times less money, because it never trains. But training is a one-time cost that amortizes: a specialist trained once and then served crosses over to being the more energy-efficient choice after roughly 2,800–4,400 forecasts on CPU, about 120–180 days of hourly 24-hour-ahead service for one series. Refresh more often than that and the zero-shot model wins on cost; train once and serve for a year and the specialist does. Because price and PUE act on the objective as linear rescalings of λ, a different local tariff slides an operator along that same axis rather than reordering the options: no cell flips at the central assumption, and at most 40% flip at the most extreme corners.

Two limits on how hard to push this. In absolute terms the electricity cost of either tier is below 10^{-3} USD per 1,000 forecasts, so the energy argument is about scale and operational simplicity rather than the utility bill; for a single city, hardware and engineering time dominate. And the E4 result removes the middle option: because the transfer recipe neither beats zero-shot on accuracy in the scarce regime nor is demonstrably equivalent to it, there is no accuracy case for the extra pipeline it requires. This is the deployment analogue of arguments that energy belongs in the evaluation objective[35,36,47]. Supplementary Table S16 sets out the procedure for re-deriving these maps for a new city.

Limitations

Seven limitations bound our claims. First, energy and hardware. CPU-side consumption on Windows is codecarbon's constant-power estimate rather than a hardware measurement (GPU energy is NVML-measured), and we benchmark a consumer GPU

and a CPU-only configuration, not a physical embedded device, so the "edge/CPU-class" energy numbers are indicative rather than a Raspberry-Pi- or Jetson-class measurement. We also did not log per-forecast latency or peak memory for the canonical runs, so deployment-critical figures beyond energy (inference latency, RAM/VRAM footprint, model storage) are reported only as the parameter count in Table 3. The measured comparison includes per-fold training for trained tiers, so the headline energy gap is specific to frequent-retraining deployment; our amortization analysis shows a once-trained specialist eventually becomes more energy-efficient per forecast, and both figures are reported. Second, contamination. Chronos-Bolt is pretrained on large public time-series collections whose exact composition we cannot fully audit, so we cannot rule out that the UCI Beijing series or ERA5/OpenAQ-derived data entered its corpus. To guard against this we re-evaluated on strictly post-2024-10 city windows (10 cities with sufficient coverage, a subset that by construction skews toward well-instrumented cities, since it requires a recent continuous record), which postdate the model's pretraining corpus[42] and cannot have been seen: the zero-shot model remained competitive with the specialist there (mean MASE 0.415 vs 0.395, Chronos-Bolt better in 6 of 10 cities, both far below the seasonal-naïve floor; Supplementary Table S14), so the PM2.5 result is not an artifact of pretraining exposure. We nonetheless treat the public Beijing 12-station set as corroborating rather than primary evidence. Third, coverage. The panel uses one sensor per city, so within-city spatial generalization rests on the Beijing 12-station check alone. Fourth, model scope. The foundation-model tier is Chronos-Bolt-small; larger TSFMs or future revisions may shift the accuracy–cost frontier, and our conclusions are explicitly frontier-relative, not model-family-absolute[32,33]. Fifth, task and comparator scope. We test hourly, 24–48 h direct multi-horizon point forecasting, with each tier at its deployed context length (Chronos-Bolt four weeks, NAS-GRU 24 h). A context-length-matched comparison (168 h and 672 h GRU lookbacks) is a natural sensitivity check; our attempts to run it exceeded the memory of the study GPU (8 GB), so we report the comparison at native contexts and flag context length as a factor we did not isolate on this hardware. Longer horizons, probabilistic scoring rules, and multivariate targets are likewise untested here. Sixth, statistics. Split-conformal exchangeability holds only approximately under temporal dependence[48,49], and with 15 scarce cities the E4 tests have limited power. Even where we establish equivalence (PM2.5 under perfect-foresight covariates), it is at a 0.05-MASE margin, and where we do not (causal PM2.5, temperature, E4) "no significant difference" remains a parity statement, not proof of interchangeability. Seventh, period heterogeneity. Each city contributes whichever window its source record supports, so the panel is not contemporaneous, and ten of the 29 cities (34% of usable hours, seven of them scarce-tier) overlap the 2020–21 pandemic period. Every comparison here is within-city and between-tier against a per-series scale, so a local regime shift rescales the problem for all tiers at once rather than favouring one; and the post-cutoff check above, which lies entirely outside that period, reproduces the result. But the exposure is correlated with the rich/scarce contrast rather than spread evenly across it, so that check bounds the concern rather than removing it (Supplementary Information).

Outlook

Two extensions follow directly. Light fine-tuning of small TSFMs on pooled regional data (the transfer recipe applied to the foundation model rather than the searched model) would test whether the two strategies' strengths combine. The decision-rule framework also generalizes: any deployment context can re-derive winner maps by inserting local electricity prices, hardware, retraining frequency, and its own cost-penalty coefficient, since all code, analysis plans, and per-claim provenance ledgers are released. As monitoring networks in low- and middle-income countries grow[4,5], our results indicate that the marginal cost of a competent forecast for a new city is now close to the cost of running a small pretrained model's inference loop, which bears directly on how air-quality early-warning capacity can be scaled[1,2].

Methods

Study design

We compare three deployment strategies for hyper-local hourly forecasting in data-scarce cities: *search* (a NAS-discovered tiny recurrent model), *specialize* (a gradient-boosting model with covariates), and *zero-shot foundation model*. With the covariate-residual foundation-model variant and a seasonal-naïve floor, these three strategies are realized as the five model tiers reported throughout. Both domains, urban PM2.5 (primary) and 2 m air temperature (secondary), use the same 29-city panel and the same evaluation harness. The study extends our earlier conference paper[24], from which it inherits the NAS-discovered architecture, the weather domain, and the edge/energy framing; the foundation-model tier, the cost-adjusted decision rule, the air-quality domain, the cross-domain generality test, and the transfer-versus-zero-shot crux experiment are new.

Data sources

PM2.5. Hourly PM2.5 from the OpenAQ v3 API[4], one sensor per city (sensor IDs in `cities_manifest.csv`). For each candidate city we selected the single OpenAQ sensor with the longest gap-gated usable window (the deterministic selection rule; ties broken by higher pre-interpolation coverage), so "reference sensor" here means the best-covered instrument available for that city rather than a certified reference-grade monitor; OpenAQ aggregates heterogeneous instruments, and we do not

assume uniform calibration across cities. Retrieval was month-windowed, resumable, and checkpointed, with per-request record-count verification (Supplementary Information). **Temperature and meteorological covariates.** The Open-Meteo historical archive[50], a gap-free reanalysis product built on ERA5[51]: 2 m temperature as the target (weather domain) plus the seven surface covariates used in our conference study: relative humidity, surface pressure, 10 m wind speed and direction, precipitation, cloud cover, and shortwave radiation. In the temperature domain the covariate set excludes the target and contains no variable mathematically derived from it. Because these are reanalysis fields rather than a single physical station, the temperature target is a location-specific gridded series; we describe it as location-level rather than station-level, and note that its "data scarcity" is imposed by clipping to each city's PM2.5 window (Section on panel construction) rather than being intrinsic to the gap-free weather record. **Beijing depth check.** The UCI Beijing Multi-Site Air Quality dataset (12 stations, hourly, March 2013–February 2017, meteorology included)[46].

Data quality gating and preprocessing

PM2.5 readings ≤ 0 or ≥ 985 µg m^{-3} were treated as missing before any gating. Non-positive concentrations are physically invalid for PM2.5 mass; 985 is a recurring sensor/API sentinel value rather than a genuine extreme episode. Both rules were fixed before the campaign, and panel membership is insensitive to the exact ceiling (Supplementary Information). From each sanitized series we extracted a usable window by splitting at gaps longer than 48 h, interpolating only gaps ≤ 6 h inside the kept segment, and requiring a final window of $\geq$ 2,160 h (90 days) with $\geq 60\%$ real (pre-interpolation) observations. All gate parameters were fixed before the campaign, not tuned post hoc. Mumbai's sub-hourly records were resampled to the hourly grid before gating. Three cities, London, Kathmandu, and Kolkata, failed those gates and were excluded (Supplementary Information). The final panel is 29 cities: 14 data-rich and 15 data-scarce (Supplementary Table S2).

Because each city contributes whichever window its source record supports, the panel is not contemporaneous: usable windows span 2016 to 2026, and ten cities overlap the 2020–21 pandemic period. We did not exclude that period, and we report the exposure and its consequences for the panel comparisons in the Supplementary Information.

The rich/scarce tier was assigned a priori, when the candidate-city list was fixed and before any data were retrieved, by the maturity of each city's regulatory air-quality monitoring infrastructure: data-rich cities are served by long-established government monitoring programmes (North America, Europe, developed Asia-Pacific, and the established networks of Santiago, Mexico City, and Bangkok), while data-scarce cities are those whose OpenAQ coverage comes mainly from recent, sparse, or low-cost/donor-funded deployments (South and Southeast Asia, Africa, and the Andean cities). The label is deliberately an infrastructure-context classification, not a function of realized record length (several scarce-tier cities end up with long usable windows, e.g. Nairobi; Supplementary Table S2), so tier membership was never revised after the data were seen.

For the temperature domain, each city's forecasting window was clipped to that same city's PM2.5 usable window, so cross-domain comparisons isolate the domain effect rather than confounding it with longer weather histories. The temperature loader does not extrapolate beyond observed coverage: windows are clipped to real weather coverage and only gaps ≤ 6 h are interpolated; a window that would require extrapolation is excluded rather than filled. An earlier loader version forward-filled beyond coverage for four cities and produced an implausible MASE; the error was identified during internal review, and all affected artifacts were regenerated under the corrected loader (deviations log).

Forecasting task and evaluation protocol

Hourly cadence; primary horizon h = 24 with direct multi-horizon forecasting (one specialist model per lead time); h = 48 as a supplementary sensitivity. Rolling-origin backtesting[44] with six folds, all evaluated on the final common test window per series, so that different training-history regimes are compared on identical test data. Accuracy is reported as MASE[52] with one fixed scale per series, namely the in-sample error of the seasonal-naïve (m = 24) forecast over the pre-test training history; the scale is shared across all regimes for that series so MASE remains comparable when training history varies. MAE, RMSE and normalized variants are recorded in every artifact.

Model tiers

Seasonal-naïve. 168-h persistence (same hour last week), the explicit floor tier. Note this weekly-persistence floor is a distinct model from the daily (m = 24) naïve whose in-sample error sets the fixed MASE denominator; the two serve different roles (a competitor tier versus a scale) and are not required to coincide. **LightGBM specialist.** One direct model per lead time[9] with lagged target, calendar features, and meteorological covariates; retrained per fold; `random_state = 42`. (A pre-campaign audit found `subsample = 0.8` set without `subsample_freq`, making bagging inert; the configuration was corrected and is documented in the hyperparameter table, Supplementary Table S3.) **NAS-GRU.** The Green-NAS-A architecture (two stacked GRU layers, 128 units) discovered by NSGA-II multi-objective search in our conference study[24,53]; trained with Adam (learning rate 10^{-3}), early stopping (patience 10, $\leq$ 50 epochs), 24-h lookback. Because training is stochastic, it was run at five seeds {42–46} and reported as mean ± sd; no best-seed selection. **Chronos-Bolt (zero-shot).** `amazon/chronos-bolt-small`[25,42] with a four-week context window and no training or fine-tuning of any kind; the

point forecast is the model's predicted mean. **Chronos-Bolt + covariates.** A residual scheme: a ridge regression on calendar and weather covariates is fit over the training history, the foundation model forecasts the residual (its temporal strength), and the covariate model's prediction for the horizon is added back. This tier consumes future covariates and is therefore subject to the same perfect-foresight/causal distinction as the specialist (below); in the causal configuration the weather covariates over the horizon are frozen at their last-known value at the forecast origin before the covariate model is evaluated.

Covariate timing (perfect foresight vs causal). We distinguish two covariate-timing regimes and fix their roles explicitly. **Primary (deployable): causal covariates.** Each meteorological covariate enters at its last value known at the forecast origin (calendar features remain future-known, since a calendar is deterministic); this is the configuration a real operator can run, and it is the one reported in Table 1, Fig. 2a, Fig. 3, and Fig. 5. **Ceiling (upper bound): perfect-foresight covariates.** Every covariate-using tier receives the meteorological covariate at the forecast *target* time (the standard `future_covariates` benchmark convention), equivalent to assuming a perfect weather forecast over the horizon; these numbers are reported only as a clearly-labelled upper bound on covariate-using-tier performance (Supplementary Table S13, Supplementary Fig. S1). The causal ablation was logged prospectively in the deviations record before it was run, and the full 29-city panel was re-run in both domains for both covariate-consuming tiers: the LightGBM specialist (the strongest covariate user and the tier carrying the domain claim) and the covariate-residual Chronos variant. NAS-GRU consumes only the past context window and is causal by construction. The univariate zero-shot tier uses no covariates and is identical in both regimes. This primary/ceiling hierarchy is stated identically in every table and figure caption that involves a covariate-using tier.

E4: transfer learning versus zero-shot (crux experiment)

NAS-GRU was pretrained on the pooled 14 rich-tier cities with per-city z-scoring (the target scarce city is never in the pretraining corpus), then fine-tuned on each of the 15 scarce-tier cities at nominal budgets of {0, 1, 10, 100}% of that city's training window, five seeds each. Comparators on identical test folds: zero-shot Chronos-Bolt, and a LightGBM specialist refit on the same nominal budget, so the comparison is "transfer versus every strategy at the same data budget," not "transfer versus nothing." Two features of the smallest budgets both cut against the transfer model's favour and are detailed in the Supplementary Information: the 0% condition uses target-city data for normalization though not for gradient updates, and the two window floors are asymmetric, so at the 1% budget the LightGBM baseline receives up to $\sim 4\times$ more data. Actual hours used are recorded per row and reported alongside nominal fractions (Supplementary Table S10). The outcome was classified against a pre-specified four-way interpretation matrix by city-level paired Wilcoxon tests, Holm-corrected across fractions, not by comparing means.

Energy and cost measurement

Primary measurement wraps each tier's entire runner call with codecarbon[45]: GPU energy is measured with NVML on the study machine's RTX 3060 Ti (the Chronos and NAS-GRU tiers), while CPU energy is codecarbon's constant-power estimate, a limitation of power instrumentation on Windows that we report as such. Energy is normalized to joules per 1,000 forecasts and converted to USD at 0.15 USD/kWh with PUE 1.4 (defaults varied in a sensitivity analysis, Supplementary Table S11). Note that for trained tiers the measured figure includes per-fold training as well as inference, whereas a supplementary TDP $\times$ latency proxy measures inference only; the two are therefore not directly comparable for trained tiers and are never averaged or ratioed against each other. Repeatability: a fixed workload was rerun five times on three cities (Beijing, Seoul, Nairobi); any city$\times$tier cell with sd/mean > 20% is reported as a range rather than a point estimate (4 of 15 cells; Supplementary Table S8)[35,47].

Cost-adjusted decision rule. For each training-history regime (4–104 weeks, city-dependent) we select the tier minimizing MASE + $\lambda \cdot$ (USD per 1,000 forecasts), where the cost-penalty coefficient λ is swept over {0, 500, 1500, 5000, 20000} MASE units per USD per 1,000 forecasts, producing winner maps per domain for Beijing (depth city), Seoul (rich tier), and Nairobi (scarce tier). Electricity price and PUE act on this objective as linear rescalings of λ; the sensitivity sweep quantifies map stability.

All timestamps are handled in Coordinated Universal Time (UTC) end to end, so calendar features are computed on the UTC clock rather than each city's local time. The same convention applies to every tier and both domains, so it cannot bias any of the contrasts reported here; the full treatment is in the Supplementary Information.

Statistical framework

Hypothesis families. The confirmatory comparisons are the two Table 1 causal-covariate domain comparisons (specialist vs zero-shot foundation model on PM2.5 and on temperature) and the four E4 fine-tune budgets (transfer vs zero-shot); city-level DM significance is FDR-controlled within each domain, and E4 Wilcoxon tests are Holm-corrected across the four budgets. All other reported P values are secondary or sensitivity analyses and are read descriptively: the perfect-foresight ceilings (Table S13), the 48-hour horizon (Table S9), the panel-level sign/Wilcoxon and Friedman–Nemenyi robustness checks, the equivalence

(TOST) analyses, and the data-volume correlations. We flag each as such where reported and do not apply cross-family multiplicity correction across these distinct question families.

Pairwise per-city comparisons use the Diebold–Mariano test[54] on absolute-error loss with the Harvey–Leybourne–Newbold small-sample correction[55]. The loss series for each city is the per-step absolute error pooled over all six folds and all 24 lead times ($\approx$ 144 error values per city, one 24-step trajectory per fold origin), so the loss series covers the same horizons that MASE averages; h = 24 enters only as the forecast-horizon order in the HLN correction, not as a lead-24-only filter. Significant-win counts are additionally controlled with Benjamini–Hochberg FDR across the 29 cities. Because a non-significant test is not evidence of equivalence, the three "tie" comparisons (PM2.5 specialist vs foundation model, causal-covariate temperature specialist vs foundation model, and E4 transfer vs zero-shot) are additionally assessed with a paired two-one-sided test (TOST) against an equivalence margin of $\delta = 0.05$ MASE, at $\alpha = 0.05$.

Two conventions are worth stating explicitly, because they put several different numbers close together. First, δ and α are distinct quantities that happen to share the value 0.05: $\delta = 0.05$ is a margin measured in MASE units, whereas $\alpha = 0.05$ is a significance level. Second, the interval TOST tests against is the $1 - 2\alpha = 90\%$ confidence interval, not the 95% one. Equivalence is declared when that 90% interval lies entirely inside the margin. This is why every TOST interval reported here is a 90% interval, while the accompanying 10,000-resample bootstrap CI on the paired difference is a conventional 95% one.

The $\delta = 0.05$ margin is deliberately strict: on PM2.5 it corresponds to roughly 0.8 $\mu g\, m^{-3}$ of MAE at the rich-tier scale, a difference an operator would consider negligible. The verdict does depend on it, so we report the full margin sensitivity (Supplementary Table S15) and state that dependence rather than presenting equivalence as margin-free. The provenance of both the equivalence analysis and the margin is documented in the Supplementary Information.

Panel-level comparisons use a binomial sign test on per-city wins and a Wilcoxon signed-rank test on paired per-city MASE; a Friedman test with Nemenyi post-hoc across all five tiers[56] is reported as a supplementary robustness check. Uncertainty is quantified with split-conformal 95% prediction intervals[57–59] calibrated on the first half of each series' backtest predictions and evaluated on the second half, pooled per tier $\times$ domain (Supplementary Table S4; per-city in Supplementary Tables S6 and S7); we note that conformal exchangeability is only approximate under temporal dependence[48,49]. The relationship between data volume and the foundation-model advantage is a Pearson correlation between usable hours and the per-city specialist-minus-FM MASE gap, with a 10,000-resample bootstrap 95% CI. Multi-seed tiers aggregate to a per-city seed-mean before any test.

Reproducibility

All canonical runs were executed under a tagged code state with per-run configuration records (`_runconfig.json`), a run registry, and SHA-256 manifests of all input data. Every quantitative claim in this manuscript is backed by a row in a results ledger mapping the claim to the artifact and exact command that produced it, and was re-verified mechanically before submission. Seeds: LightGBM 42; NAS-GRU {42–46}. The OpenAQ API key is supplied through an environment variable and never stored in code, logs, or history. Code, the analysis plan, deviations log, and fetch scripts are released with the paper.

Data availability

The datasets analysed during the current study are available in the OpenAQ repository, https://openaq.org (hourly PM2.5; sensor IDs for all 29 cities are listed in Supplementary Table S2), the Open-Meteo historical archive, https://open-meteo.com (temperature and meteorological covariates), and the UCI Machine Learning Repository, https://archive.ics.uci.edu/dataset/501/beijing+multi+site+air+quality+data (Beijing multi-site air-quality data). The scripts that reproduce the exact extracts used in this study, together with SHA-256 manifests of every input file, are available in the code repository referenced in the Code availability statement below.

Code availability

The full harness, analysis scripts, the analysis plan with deviations log, the per-claim results ledger, and figure/table generation code are available at https://github.com/Muhtasim-Munif-Fahim/cost-aware-tsfm-forecasting under the tagged release `v1.0-campaign` used for the canonical campaign.

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Author contributions

M.M.M.F.: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data Curation, Visualization, Writing – Original Draft. M.R.K.: Conceptualization, Supervision, Writing – Review & Editing. Both authors reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

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